포트폴리오

제형과 공정 이해를 바탕으로 역량을 쌓아온 **김지연**입니다

[지원자]

김지연

🎂 1999.11.18
☎ 010-5350-3948
✉ delayeon18@naver.com

성장하는 연구원 **김지연**입니다

제형 및 공정 최적화와 품질 혁신으로
환자의 삶에 기여하겠습니다



Profile

김지연 | Kim Ji Yeon

1999.11.18
delayeon18@naver.com

Education

박사	대구가톨릭대학교 약학과 수료 2023.03 – 2025.02
석사	인제대학교 산업약제학 전공 졸업 2021.09 – 2023.02
학사	인제대학교 제약공학과 졸업 2018.03 – 2021.08

Career

케이아이바이오 ㈜ | KI Bio Co., Ltd.
연구개발전담부서 | R&D center
2025.03 – 2025.12

" Pre-formulation 부터 처방 및 공정 최적화, PAT & AI 기반 품질예측까지 의약품 개발 전주기에서 **데이터 기반의 문제 해결과 성과 창출** 경험 보유 "

Skill & Competency

제형 및 공정 설계(QbD)

- Pre-formulation 연구 및 **다양한 제형 (고형제, 패치, 렌즈 등) 최적화**
- DoE 기반 품질 영향 변수 도출 및 최적 조건 확립

≫ HPLC, UV-Vis, DoE, Monte Carlo simulation 등 활용

PAT & 실시간 품질 관리

- NIR, 이미지 등 분석 기기를 활용한 **제조 공정의 실시간 모니터링**
- 다변량 분석(PCA, PLS)을 통한 **공정 변동성 예측 및 품질 관리 전략 수립**

≫ NIR, FT-IR, 초고속 카메라, 초분광 카메라, PCA, PLS 등 활용

AI 기반 품질 평가/예측

- **Deep Learning 기반** 이미지 분석 모델 및 예측 모델을 활용한 품질평가
- Pharma 4.0 대응 **데이터 기반 의사결정 및 품질 예측**

≫ MATLAB, Python, R 활용

기술이전 및 CTD 문서화

- 수학적 모델링 기반 **scale-up 기술이전 지원** 및 공정 타당성 검토 경험
- CTD 3.2.P 섹션 작성 및 결과 보고서 작성

≫ 메커니즘 기반 수학적 모델링, 시뮬레이션 활용

Publications

2021–2026 | SCI(E) 논문 10편 (1저자 9편)

연도	게재지/논문 제목	역할
2026	Eur. J. Pharm. Sci. | **Development and optimization of an LED–based particle image velocimetry methodology for dynamic powder flowability in pharmaceutical manufacturing**	1저자
2025	J. Pharm. Investig. | **Integrated control strategy based on deep learning and process analytical technology for the manufacture of a triple fixed-dose combination tablet**	1저자
2025	Int. J. Pharm. | **Deep learning-based image classification and quantification models for tablet sticking**	1저자

R&D Projects

2020–2025 | 국가 R&D 연구과제 10건 참여 | 국내 제약회사 용역과제 참여

기관	과제명	참여 기간
산업통상자원부	고품질 설계기반 고혈압 치료제 생산용 핵심 소재 및 중간체, 완제 생산 공정 기술개발	25.03 - 25.12
산업통상자원부	항체의약품의 장내 수송체 표적형 경구 약물전달시스템 및 제형화 기술개발	24.06 - 25.02
산업통상자원부	안구건조증 치료 약물(lifitegrast)의 흡수율을 증가시키기 위한 점안렌즈를 이용한 약물전달 응용기술 개발	20.04 - 24.12

CONTENT.1

Master's thesis

석사 졸업논문

01

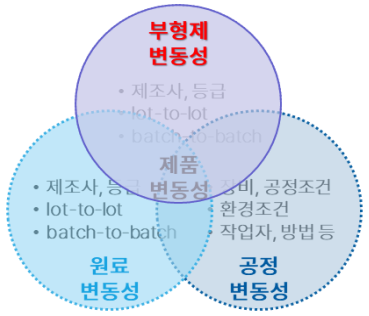
- ▶ 연구배경 및 필요성
- ▶ 연구 방법
- ▶ 주요 연구결과

연구배경 및 필요성

의약품 설계기반 품질고도화 접근법을 이용한 의약품 개발과정에서 부형제의 변동성이 품질에 미치는 영향과 관리전략 개발

Development control strategy for excipient variability in quality by design approach

01 완제의약품 변동성

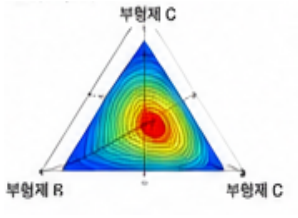


$\sigma^2_{제품} = \sigma^2_{원료} + \sigma^2_{공정} + \sigma^2_{부형제} + \sigma^2_{상호작용}$

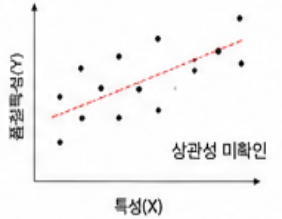
- 완제의약품 변동성은 **부형제, 원료, 공정 변동성 및 이들의 상호작용**으로 인해 발생
- 특히 **부형제의 물리화학적 특성(입도분포 수분함량, 밀도 등)**은 품질특성에 직접적인 영향을 미치며, 제조사/등급, lot, batch 별로 달리 나타날 수 있음

02 기존 연구의 한계

- 기존 연구는 주로 부형제 양/농도 변화에 따른 품질 변동만 확인
- 설계공간 내에서 부형제 변동이 미치는 영향에 대한 연구 부족



- 품질변동을 야기한 물리화학적 특성과 품질특성 간의 상관관계 연구 부족
- 변수 간 상관관계 규명 미흡



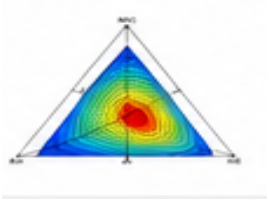
- 정확한 원인 파악 불가로 **효과적인 관리전략 수립 어려움**
- 지속적 품질 보장을 위한 과학적 관리 방안 부족



03 본 연구의 목적 및 주요 내용

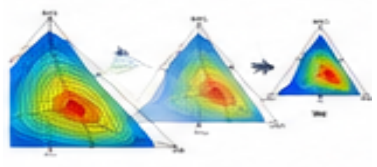
(1) 설계공간 개발

- D-optimal 혼합설계
- 최적 처방 도출
- 설계공간 확립



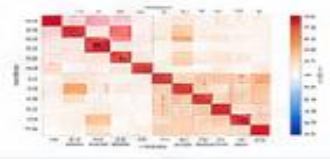
(2) 부형제 변동성 평가

- 다양한 제조사. 등급의 부형제 (MCC, PVP) 사용
- 설계공간의 위치 이동 및 크기 변화 평가



(3) 원인 규명 및 상관관계 분석

- 부형제 물리화학적 특성 측정
- PLS로 주요 영향 특성 도출
- 피어슨 상관계수로 물리화학적-품질특성 상관관계 확인



(4) 예측 모델 개발

- 물리화학적 특성을 input으로 한 용출 예측 모델 개발
- 예측 성능 평가(RMSE, %RE 등)



(5) 관리전략 제시

- 설계공간 내 운영을 위한 관리전략 제시
- 지속적 품질보장을 위한 관리전략 제안

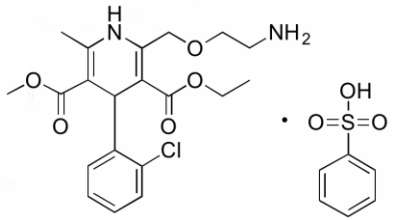


연구 방법

의약품 설계기반 품질고도화 접근법을 이용한 의약품 개발과정에서 부형제의 변동성이 품질에 미치는 영향과 관리전략 개발

Development control strategy for excipient variability in quality by design approach

① 모델 약물 및 제형



Amlodipine besylate
(BCS Class I)

주요 부형제



MCC (부형제)

PVP (결합제)

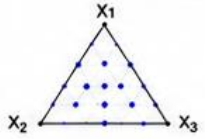
② QbD 기반 제형 개발
(설계공간 구축)

위험평가 (FMEA)



- 위험요인 도출
- CMP/CMA 선정

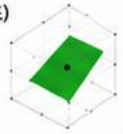
실험계획법 (DoE)
(D-optimal Mixture Design)



- 조성비 범위 설정
- 실험 및 모델링

최적화 및 설계공간 구축

- 목표: QTPP (용출/함량/경도/마손도)
- 반응 최적화
- 설계공간(Design space) 확립



③ 부형제 변동성 평가
(설계공간 내 평가)

MCC 변동성 평가

	A사	B사	C사	…
등급/lot	G1	G2	G3	…

PVP 변동성 평가

	A사	B사	C사	…
등급/lot	K12	K17	K25	…

설계공간 내 조성으로
각 부형제 조합별 tablet 제조



④ 품질 특성 평가
(CQAs)



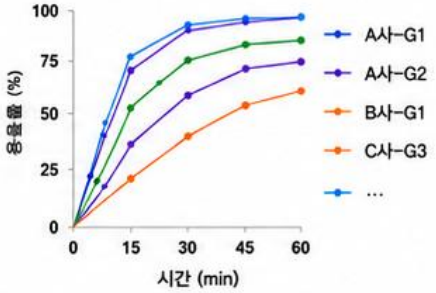
경도 마손도 용출시험



함량 (Assay)

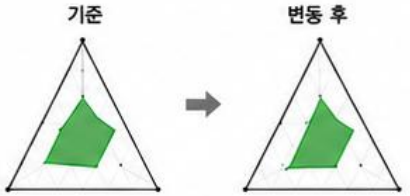


함량 균임성 (Content Uniformity)

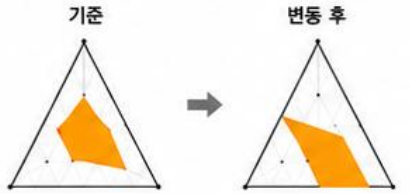


⑤ 설계공간 변동성 확인
(제조사/등급에 의한 영향)

MCC 변동 시 설계공간 이동



PVP 변동 시 설계공간 이동



- ✓ 설계공간의 크기/위치 변화 평가
- ✓ 품질 목표 달성 여부 확인

6. 원인 규명 및 예측 모델 개발

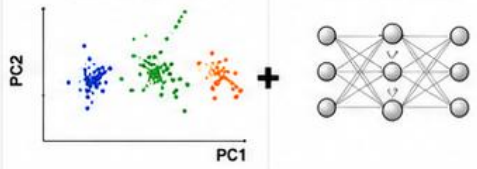
6-1. 부형제의 물리화학적 특성 평가



- 수분함량
- 입도분포, Hausner ratio
- 압축지수, 탭밀도
- K value, PSD 등

6-2. 통계 분석 (원인 규명)

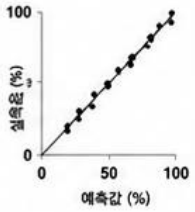
PCA (주성분 분석) ANN (인공신경망)



- 주요 영향 요인 파악
- 품질특성(CQAs) 예측 모델 구축

6-3. 모델 검증

- 교차검증 (K-fold CV)
- 지표: RMSE, MAE, MAPE, R^2 등
- 실측값 vs 예측값 비교 (Bland-Altman)



7. 관리 전략 개발 및 적용

제어전략 수립
원료 사양 설정 및 관리, 공정 파라미터 관리

실시간 모니터링
물리화학적 특성 및 CQAs 지속 모니터링

지속적 개선 (Continual Improvement)
모델 업데이트 및 설계공간 재평가

일관된
고품질 속방정
지속적 생산



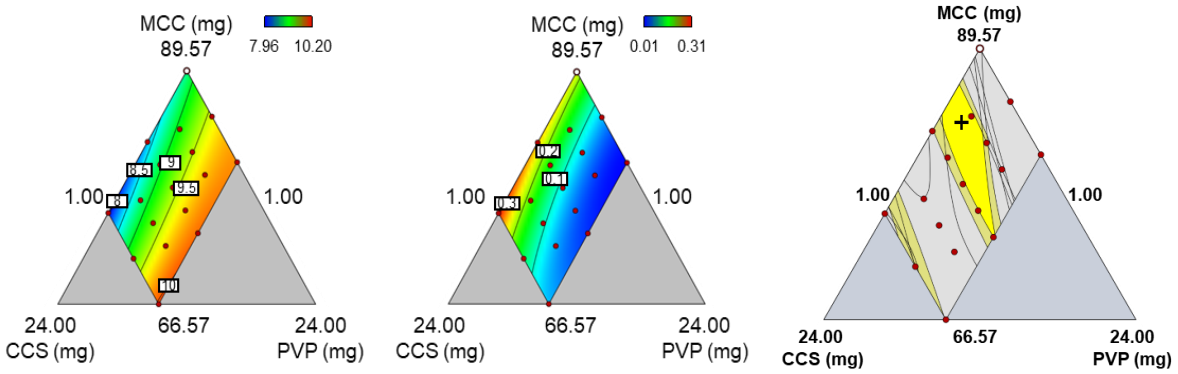
주요 연구결과

의약품 설계기반 품질고도화 접근법을 이용한 의약품 개발과정에서 부형제의 변동성이 품질에 미치는 영향과 관리전략 개발

Development control strategy for excipient variability in quality by design approach

(1) DoE를 통한 처방 최적화

- D-optimal mixture design을 사용한 실험수행
- 설계공간(Design space) 도출



(2) MCC 및 PVP의 주요 물리화학적 특성 도출

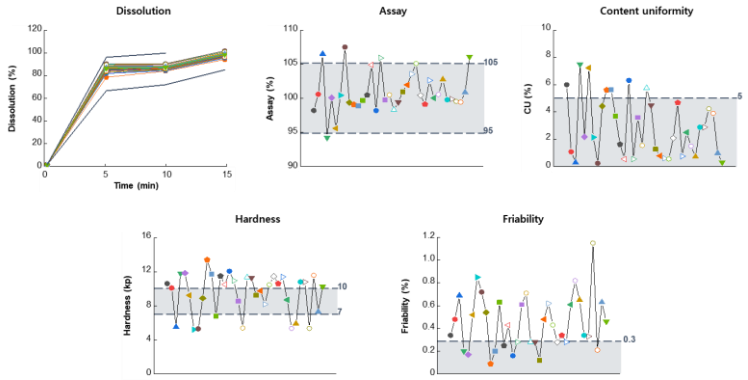
- Risk assessment를 통한 주요 물리화학적 특성 도출

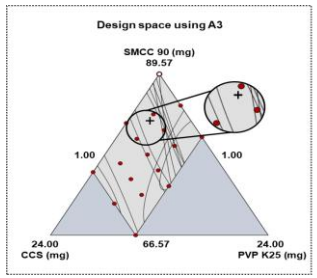
Attributes	CQAs				
	Assay	CU	Dissolution	Hardness	Friability
Appearance	Low	Low	Low	Low	Low
Solubility	Low	Low	Medium	Low	Low
Identification A (IR)	Low	Low	Low	Low	Low
Identification B (Wet chemistry)	Low	Low	Low	Low	Low
Identification C (Degree of polymerization)	Medium	Medium	Medium	Medium	Medium
Residue on ignition (Sulfated ash)	Low	Low	Low	Low	Low
Residual solvent	Low	Low	Low	Low	Low
Solubility (Ammoniacal solution of copper tetramine R)	Low	Low	Low	Low	Low
Conductivity	Low	Low	Low	Low	Low
pH	Low	Low	Medium	Low	Low
Loss on drying	Medium	Medium	High	High	High
Bulk density	High	High	High	High	High
Tapped density	High	High	High	High	High
True density	Low	Low	Low	Medium	Medium
Particle size	High	High	High	High	High
Particle size distribution	High	High	High	High	High
Powder flow	Medium	Medium	Medium	Medium	Medium
Water-soluble substances	Low	Low	Low	Low	Low
Ether-soluble substances	Low	Low	Low	Low	Low
Heavy metals	Low	Low	Low	Low	Low
Mercury	Low	Low	Low	Low	Low
Cadmium	Low	Low	Low	Low	Low
Lead	Low	Low	Low	Low	Low
Arsenic	Low	Low	Low	Low	Low
Starch	Low	Low	Low	Low	Low
Total aerobic microbial count	Low	Low	Low	Low	Low
Total combined molds and yeasts count	Low	Low	Low	Low	Low
Absence of Staphylococcus aureus	Low	Low	Low	Low	Low
Absence of Pseudomonas aeruginosa	Low	Low	Low	Low	Low
Absence of Escherichia coli	Low	Low	Low	Low	Low
Absence of Salmonella species	Low	Low	Low	Low	Low
Absence of Enterobacteriaceae	Low	Low	Low	Low	Low
Absence of Coliform species	Low	Low	Low	Low	Low
Packaging and Storage	Low	Low	Low	Low	Low

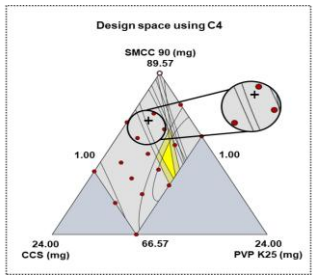
Attributes	CQAs				
	Assay	CU	Dissolution	Hardness	Friability
Identification	Low	Low	Low	Low	Low
Nitrogen	Low	Low	Low	Low	Low
Residue on ignition	Low	Low	Low	Low	Low
Lead	Low	Low	Low	Low	Low
Limit of aldehydes	Low	Low	Low	Low	Low
Limit of hydrazine	Low	Low	Low	Low	Low
Vinyl pyrrolidinone	Low	Low	Low	Low	Low
2-pyrrolidone	Low	Low	Low	Low	Low
Peroxides	Low	Low	Low	Low	Low
Formic acid	Low	Low	Low	Low	Low
pH	Low	Low	Medium	Low	Low
Moisture	Medium	Medium	High	High	High
K value	Medium	Medium	High	High	High
PSD	High	High	High	High	High
Bulk density	High	High	High	High	High
Tapped density	High	High	High	High	High
Flowability	Medium	Medium	Medium	Medium	Medium
Packaging and storage	Low	Low	Low	Low	Low

(3) 제조사/등급에 따른 설계공간 변동 확인

- 제조사/등급이 다른 MCC 및 PVP를 사용한 완제의약품 품질평가 후 설계공간 이동 확인

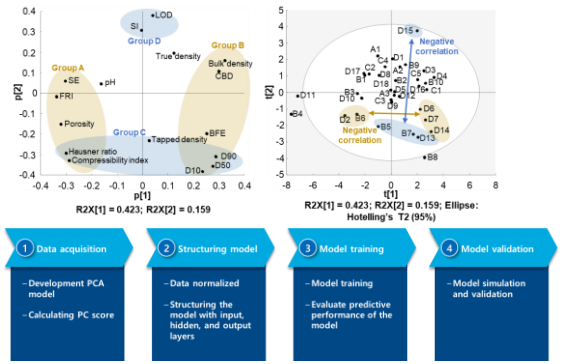


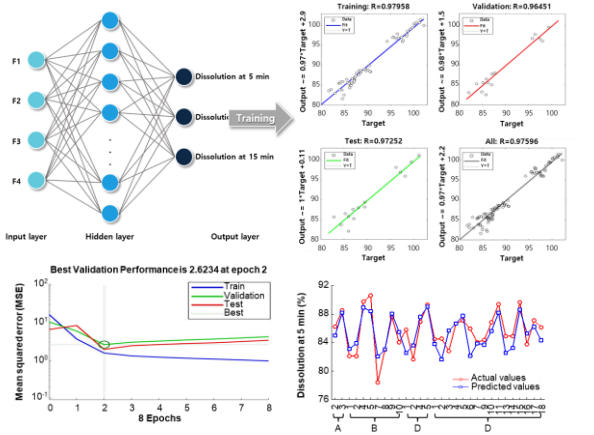




(4) 완제의약품 품질 예측 모델 개발

- 다변량 분석을 통한 물리화학적 특성-완제품 품질특성 상관관계 확인 후 제조사/등급 변동에 따른 완제의약품 품질 예측 모델 개발





CONTENT.2

Publications

SCI(E) 논문

02

- ▶ 정제 스티킹 평가를 위한 딥러닝 기반 영상 분류 및 정량 분석 모델
- ▶ 이중 방출형 이중층 정제의 제형 개발 전략: QbD와 위약층의 통합적 접근법

논문 요약

정제 스티킹 평가를 위한 딥러닝 기반 영상 분류 및 정량 분석 모델

Deep learning-based image classification and quantification models for tablet sticking

International Journal of Pharmaceutics, 678, 125690 (2025) | 1저자

Deep Learning | Image Analysis | MATLAB | Python | R

논문 요약

- **최적의 CNN 모델 선정:** AlexNet, VGG 16, ResNet 50, GoogLeNet 중 정확도(99.39%)와 효율성이 가장 높은 GoogLeNet을 분류 모델로 채택
- **텍스처 기반 정량화:** 육안으로 확인하기 어려운 미세한 질감 변화를 GLCM 특징으로 추출하여 스티킹 지수(SI) 계산

역할

- **데이터 전처리 및 증강:** 밝기 조절, 회전, 노이즈 추가 등을 통해 모델의 견고함을 높이기 위한 학습 데이터셋 구축
- **분류 모델 성능 평가:** 4가지 CNN 아키텍처별 정확도, 정밀도, 재현율, F1-score 산출 및 비교분석
- **정량화 알고리즘 검증:** 실제 제조 환경에서 생산된 10개 배치의 정제 이미지를 활용한 스티킹 지수(SI) 산출 및 모델의 반복성 검증



International Journal of Pharmaceutics 678 (2025) 125690

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journal homepage: www.elsevier.com/locate/ijpharm

Deep learning-based image classification and quantification models for tablet sticking

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ARTICLE INFO

Keywords:
Sticking
Image analysis
Deep learning
Convolutional neural network
Support vector machines
Visual inspection

ABSTRACT

Sticking can significantly affect drug product quality, manufacturing efficiency, and therapeutic efficacy in pharmaceutical tablet manufacturing. This study presents a novel integrated model with a convolutional neural network (CNN) and gray-level co-occurrence matrix (GLCM) based features combined with a support vector machine to classify and quantify tablet sticking. The classification model was developed and evaluated using CNN architectures, including AlexNet, VGG 16, ResNet 50, and GoogLeNet. GoogLeNet showed the best performance in terms of accuracy (99.39 %), precision (100.00 %), recall (98.78 %), F1-score (99.38 %), and computational efficiency. GLCM features such as energy, homogeneity, contrast, and correlation were analyzed to develop an optimal quantification model, revealing a significant difference between the sticking and non-sticking regions. Based on these differences, the sticking regions were detected and quantified using a sticking index. To validate the final model, which integrated the classification and quantification models, 10 batches of tablets were produced using a rotary tablet press. The validation confirmed high measurement repeatability with minimal and classified sticking levels. Tablet quality attributes such as assay, content uniformity, and weight were evaluated. Despite the occurrence of sticking, the tablet quality attributes met their criteria. These results suggest that measuring tablet quality attributes and visual inspection may not be sufficient to detect mild sticking. However, the integrated model proposed in this study could detect mild sticking, even if the tablet quality attributes remained within the acceptable criteria. This study demonstrated that the proposed integrated model could improve pharmaceutical manufacturing efficiency, ensure consistent drug product quality, and overcome visual inspection limitations.

1. Introduction

Generally, sticking may occur when tablet components adhere to the punch surfaces or die walls during compression. The factors influencing sticking include the adhesion of the active pharmaceutical ingredient (API), its interaction with excipients, and the surface characteristics of the tooling material (i.e., punch and die) (Gunawardana et al., 2023, Patel et al., 2006). Moreover, inadequate lubrication, high moisture content, and inappropriate particle size distribution accelerate sticking, making the manufacturing process more complex. Sticking is a significant defect that affects drug product quality and manufacturing efficiency, leading to patient safety risks, production delays, and additional costs (Gunawardana et al., 2023, Paul et al., 2017). Therefore, it is necessary to prevent and manage sticking to obtain high-quality products and improve manufacturing efficiency. Although sticking is a significant phenomenon in the pharmaceutical industry, there are no standardized regulatory methods for precisely observing and quantitatively assessing it. One of the most popular methods to detect sticking is visual inspection; however, this increases the variability of results owing to its dependence on the inspectors' subjective judgment. In addition, visual inspection has limitations in detecting mild sticking at an early stage or quantifying its severity. Advanced techniques such as atomic force microscopy analysis (Wang et al., 2003), fluid dynamics analysis (Christodoulou et al., 2020), and force–displacement measurements (Bunker et al., 2011) have been applied to overcome these limitations. However, these methods are primarily applied at the laboratory scale and are unsuitable for real-time monitoring in pharmaceutical manufacturing environments.

Recently, machine learning, particularly deep learning-based analysis, has been widely used as a powerful tool for visual defect detection across various industries (Hütten et al., 2024, Lin et al., 2024, Saberironaghi et al., 2023). Deep learning models can process large volumes

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Available online 6 May 2025
0378-5173/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

연구 방법

정제 스티킹 평가를 위한 딥러닝 기반 영상 분류 및 정량 분석 모델

I. 분류 모델 (Sticking 여부 분류)

1. 데이터 수집 및 증강

- 밝기 조절
- 좌/우, 상/하 반전
- 대비 조절
- 회전 변환
- 크기 변환
- 노이즈 추가

2. CNN 모델 학습

- AlexNet
- VGG 16
- GoogLeNet
- ResNet 50

합성곱 정규화 활성화 풀링 완전연결 드롭아웃

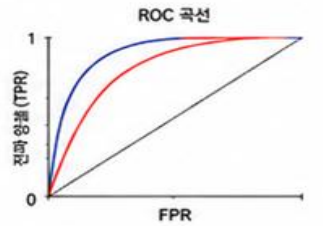
3. 모델 평가

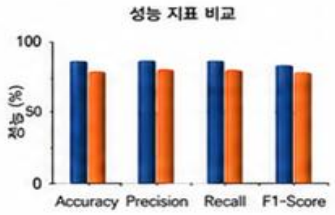
정확도 (Accuracy)
= (TP + TN) / (TP + TN + FP + FN)

정밀도 (Precision)
= TP / (TP + FP)

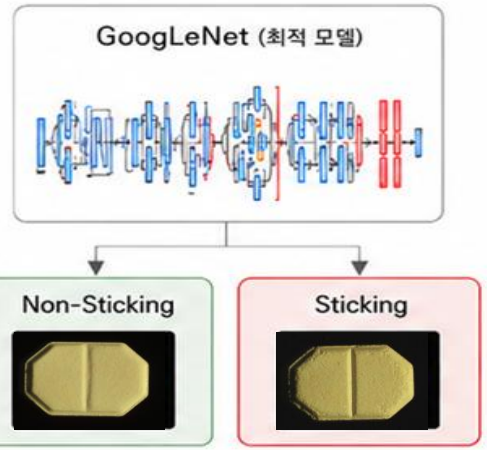
재현율 (Recall)
= TP / (TP + FN)

F1-Score
= 2 × (Precision × Recall) / (Precision + Recall)





4. 최적 모델 선정 및 분류



II. 정량 모델 (Sticking 정도 정량화)

1. 이미지 전처리

- 그레이스케일 변환
- CLAHE 적용
- 배경 제거

2. GLCM 특징 추출 및 분석

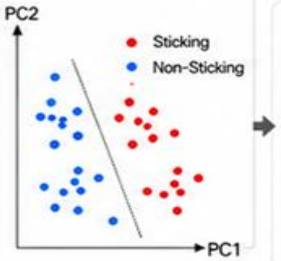
0	2	1	0
2	1	2	1
0	1	2	2
1	0	1	2

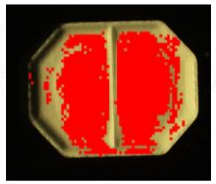
에너지 동질성 대비 상관관계 ···

PCA 수행으로 주요 성분 추출
→ 차원 축소 및 정보 손실 최소화

3. Sticking 영역 검출 및 정량

SVM 모델 학습 → Sticking 영역 검출 → Sticking 지수(SI) 계산





$$SI(\%) = \frac{\text{Sticking 픽셀 수}}{\text{전체 정제 픽셀 수}} \times 100$$

SI 값이 클수록
Sticking 정도가 높음

4. 모델 검증

① 신규 정제 이미지 입력



② Sticking 지수(SI) 계산

0.12%	2.35%	5.67%	···

③ 검증 및 성능 평가

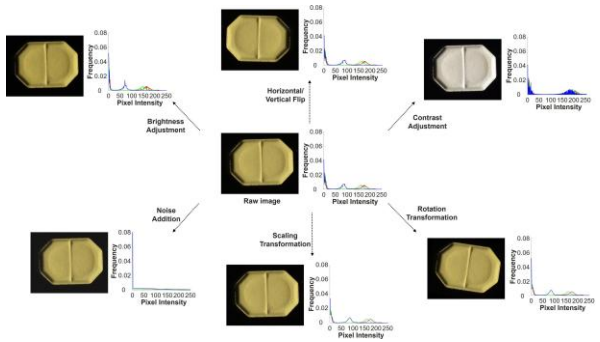
- 반복성 평가
- 재현성 평가
- 정확도 평가

주요결과 요약

정제 스티킹 평가를 위한 딥러닝 기반 영상 분류 및 정량 분석 모델

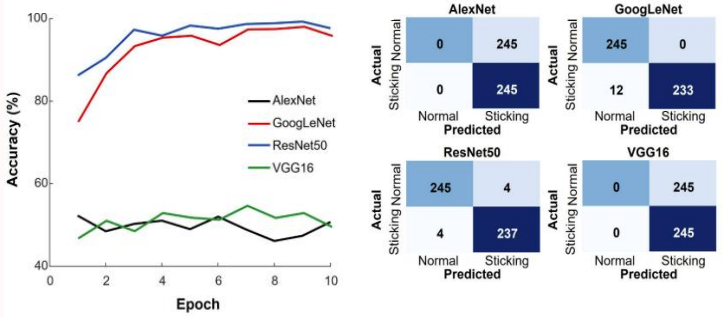
(1) Sticking 분류 모델 개발

- 데이터셋의 다양성을 높이고 다양한 조건에서 모델 성능을 개선을 위한 데이터 증강



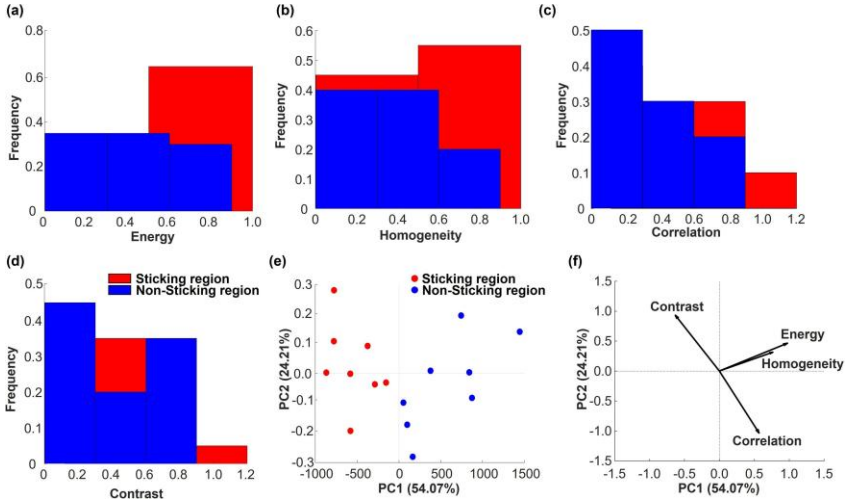
- CNN 모델에 대한 정확도/confusion matrix 평가

  ≫ 최종 모델로서 GoogLeNet 선정

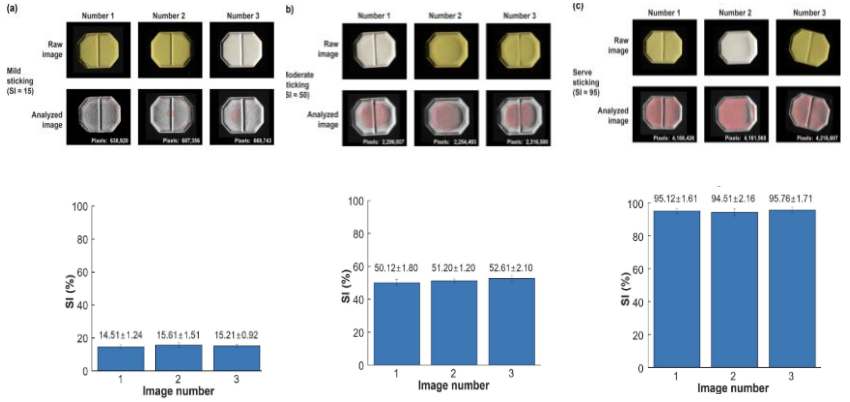


(2) Sticking 정량 모델 개발

- Sticking 및 non-sticking 영역에 대한 GLCM 특징 추출

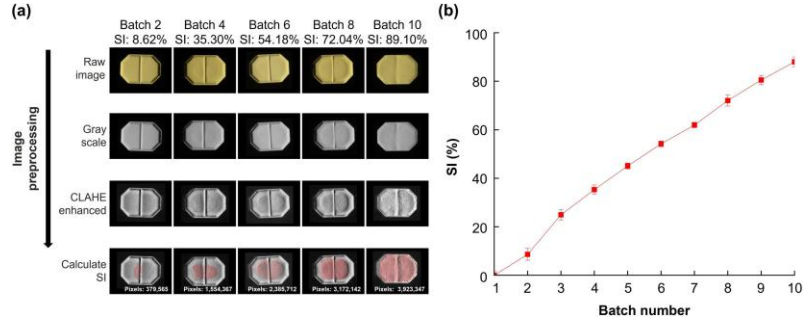


- Sticking 정도(Mild, moderate, serve)에 따른 평가

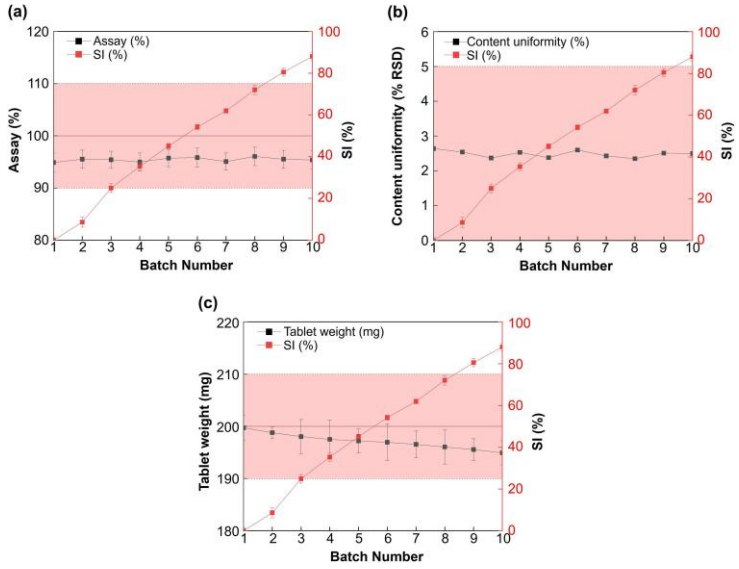


(3) 분류/정량 통합 모델 검증

- 통합 모델 검증(배치 number에 따른 SI 값 계산)



- 정제 품질특성과 SI 값 비교



논문 요약

이중 방출형 이중층 정제의 제형 개발 전략: QbD와 위약층의 통합적 접근법

A formulation development strategy for dual-release bilayer tablets: An integrated approach of quality by design and a placebo layer *International Journal of Pharmaceutics , 618 (2022)* | 1저자

QbD | Placebo Layer | Bilayer tablet | Multivariate analysis

논문 요약

- **처방 최적화:** D-optimal mixture design을 활용하여 IR 및 SR층에 대한 최적 처방 도출
- **위약층 기계적 특성 평가:** API를 대체할 수 있는 위약층 개발을 위해 위약층의 기계적 특성(붕해, 접촉각, 겔 강도, 수분함량 등)을 평가함
- **유사성 인자 계산:** API와 유사한 기계적 특성을 나타내는 첨가제 선정을 위해 ANCOVA 분석 수행
- **생물학적 동등성 검증**: 비글견을 이용한 in vivo 약동학 시험 결과, 최적화된 시험약이 대조약(Betmiga®)과 생체이용률 면에서 동등함을 확인

역할

- 위약층 스크리닝 및 기계적 특성 평가
- 통계 모델 적합 및 설계공간 확립
- 다변량 분석(MVA)을 통한 상호작용 규명
- 대조약과의 용출 유사성 시험



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journal homepage: www.elsevier.com/locate/ijpharm

A formulation development strategy for dual-release bilayer tablets: An integrated approach of quality by design and a placebo layer

Jong Kwon Han [a,1], Ji Yeon Kim [b,1], Du Hyung Choi [b,*], Eun-Seok Park [a,*]

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ARTICLE INFO

Keywords:
Quality by design
Design space
Dual-release mechanism bilayer tablet
Optimization
Development strategy

ABSTRACT

Although dual-release mechanism bilayer tablets containing one drug in both immediate- and sustained-release layers are widely used to improve therapeutic efficiency, studies quantitatively analyzing the drug amount released from each layer and the mutual effect of each layer's mechanical properties on drug product quality are limited. Here, the formulation of a dual-release bilayer tablet containing sarpogrelate HCl was optimized with a placebo layer and quality by design (QbD) approach. The placebo layer was developed to replace the active pharmaceutical ingredient and its mechanical properties were evaluated. The formulation was developed using the placebo layer to quantitatively analyze the drug released from each layer. The mixture design and Monte Carlo simulation enabled robust design space identification. The mutual effect of each layer's mechanical properties on drug product quality was confirmed by multivariate analysis using the optimal settings in the design space. The optimized formulation was characterized by comparison with a reference drug for various quality attributes and *in vivo* pharmacokinetic parameters, which ensured the bioequivalence of the optimized bilayer tablet with the reference drug. This study shows that the integration of QbD and a placebo layer is an effective optimization strategy for dual-release bilayer tablets containing one drug in different layers.

1. Introduction

Sarpogrelate hydrochloride (HCl) is used to treat coronary artery disease, angina pectoris, and atherosclerosis as an antithrombotic drug. The half-life of sarpogrelate HCl is about 0.7 h in healthy subjects (Kim et al., 2014). Because of its short half-life, the immediate-release (IR) formulation of sarpogrelate HCl is administered at 100 mg three times a day (Kim et al., 2015). However, taking drugs several times per day has poor patient compliance. Therefore, studies have attempted to develop sarpogrelate HCl as a matrix tablet for sustained-release (SR) (Kim et al., 2015); however, SR systems cannot reach minimum therapeutic systemic levels quickly. To overcome the problems mentioned, many studies have developed multilayered tablets, such as bilayer tablets and three-layered tablets that have a dual-release mechanism (Jun et al., 2018; Lee et al., 2019). Generally, dual-release mechanism bilayer tablets are designed with an IR layer for fast onset time and a controlled release layer for long duration of action (Jun et al., 2018). Bilayer tablets with a dual-release mechanism can be designed with the rationale that the minimum therapeutic sarpogrelate HCl level should be rapidly attained with the initial burst in the IR layer, which is maintained by the constant SR of sarpogrelate HCl over a defined period in the SR layer (Jun et al., 2018). Based on the reference drug (Sapodifil® SR, Alvogen, Pine Brook, NJ, USA), the bilayer tablet was composed of 100 and 200 mg of sarpogrelate HCl in the IR and SR layers, respectively. However, the high ratio of drug content can affect the mechanical properties of each layer, which could significantly affect the quality of the drug product (Ticehurst and Marziano, 2015). Particularly, in cases in which bilayer tablets contain different active pharmaceutical ingredients (APIs) in different layers, the amount of drug released from each layer can be quantitatively analyzed using the dissolution test. However, it might be challenging to quantitatively analyze the amount of drug released from each layer in bilayer tablets containing the same API in both layers.

To develop a formulation for dual-release mechanism bilayer tablets, the physical properties of the material and mechanical properties of these tablets should be evaluated. This is because the high ratio of material in each layer might significantly influence the mechanical properties of the tablet, and the mechanical properties of each layer could

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E-mail addresses: choidh@inje.ac.kr (D.H. Choi), espark@skku.edu (E.-S. Park).
[1] These two authors equally contributed to this study.

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연구 방법

이중 방출형 이중층 정제의 제형 개발 전략: QbD와 위약층의 통합적 접근법

1. 플라시보 층 개발 및 특성 평가

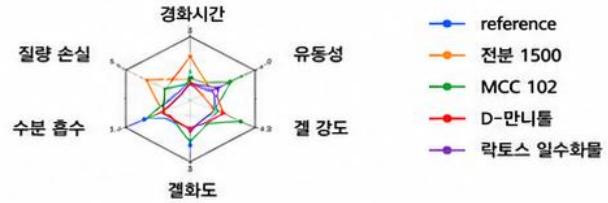
1-1. 플라시보 층 구성 성분



1-2. 주요 기계적 특성 평가

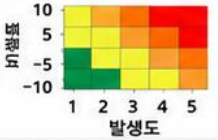


1-3. 유사성 평가



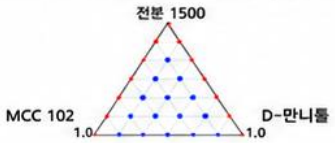
2. QbD 접근을 통한 이중층 정제 설계

2-1. 위해평가 (Risk Assessment)

CQA 정의 (품질 목표)	영향 인자 도출 (예시)	위해 수준 평가 (FMEA)
• 용출 프로파일 • 정제의 강도 • 파쇄 강도 등	• SR 층: 전분 1500, MCC 102, D-만니톨 • IR 층: 락토스, 사프그헬레이트 HCl • 압축력, 충전 질량 등	

2-2. 실험계획 (DoE)

혼합물 설계 (Mixture Design)

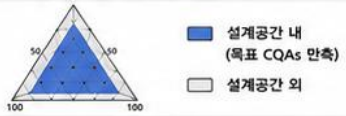


실험 조건

- 각 층의 CMAs 농도/함량 변화
- 압축력 수준 설정

2-3. 설계공간 도출 (Design Space)

혼합물 설계의 설계공간

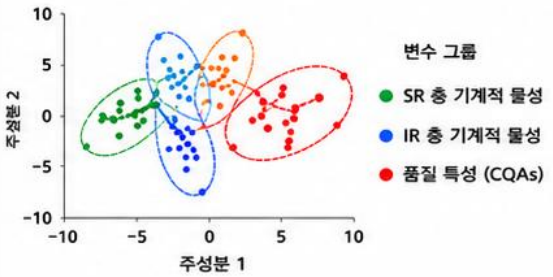


설계공간 내 목표 CQAs 예측 범위

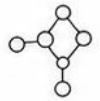


3. 상관관계 분석 (Multivariate Analysis)

CQA와 CMA 간 상관관계 분석



주요 분석 결과



기계적 물성 간 상관관계 및 영향도 규명



각 층 조성이 CQAs에 미치는 영향 확인



제형 설계 시 고려해야 할 핵심 인자 도출

4. 대조약과의 *in vitro* 평가 및 유사성 인자 (f_2) 값 도출

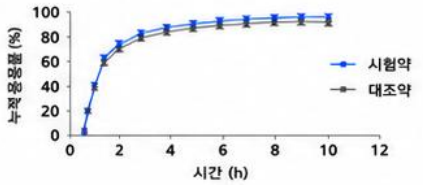
4-1. *in vitro* 용출 시험



시험 조건

- 용출액: pH 1.2 → pH 6.8
- 회전 속도: 50 rpm
- 온도: 37 ± 0.5 °C

4-2. 유사성 인자 (f_2) 계산



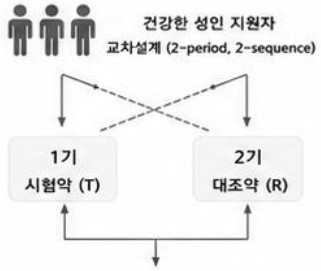
$$f_2 = 50 \times \log\left\{1 + \frac{1}{n}\sum_{t=1}^{n}(R_t - T_t)^2\right\}^{-0.5} \times 100$$

n: 측정 시점 수, R_t: 대조약 용출률, T_t: 시험약 용출률

$f_2 \geq 50$: 유사성 만족

5. 생물학적 동등성 평가를 위한 생동성 시험

5-1. 시험 설계



충분한 Wash-out 기간

5-2. 혈중 농도 분석

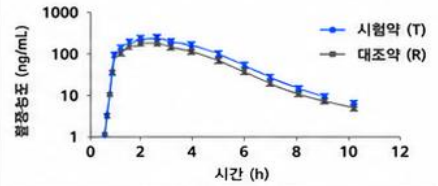


LC-MS/MS 분석
(혈장 중 사프그헬레이트 HCl)

약동학 파라미터

- C_{max}, T_{max}
- AUC_{0-t}, $AUC_{0-\infty}$
- $t_{1/2}$

5-3. 생동성 평가



통계 분석 (90% 신뢰구간)

- AUC_{0-t}, $AUC_{0-\infty}$, C_{max}
- 범위: 80.00% ~ 125.00%

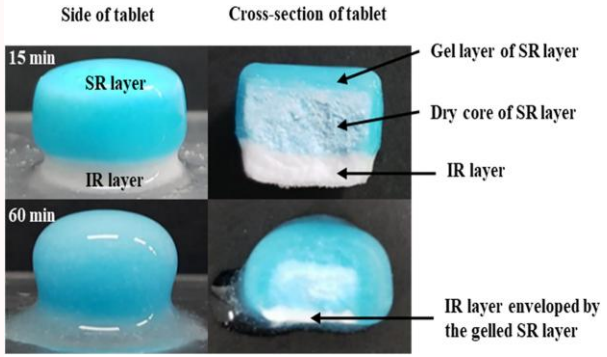
생동성 기준 충족 시 동등성 입증

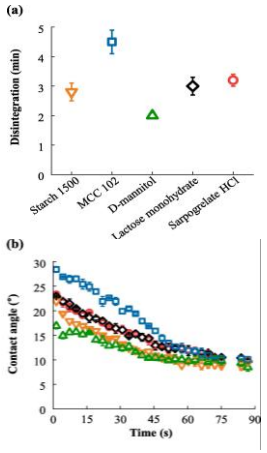
주요결과 요약

이중 방출형 이중층 정제의 제형 개발 전략: QbD와 위약층의 통합적 접근법

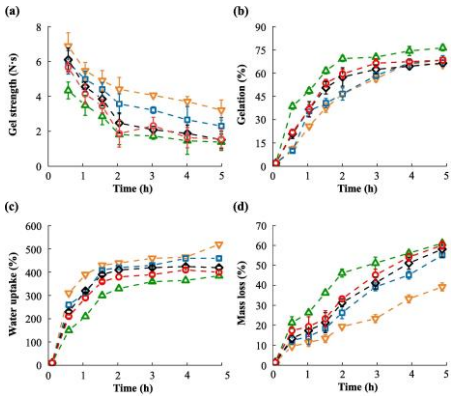
(1) 위약층 개발을 위한 기계적 특성 평가

• IR 위약층 특성평가(붕해 및 접촉각)



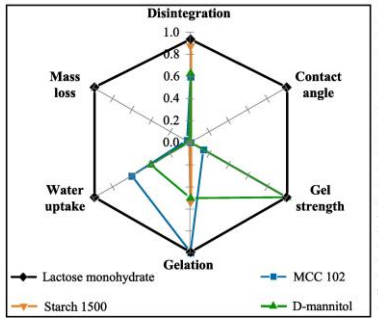


• SR 위약층 특성평가(겔강도, 겔화도, 수분함량, 질량손실)



• ANCOVA 분석 기반 유사성 인자 계산

Similarity values



Analysis of covariance

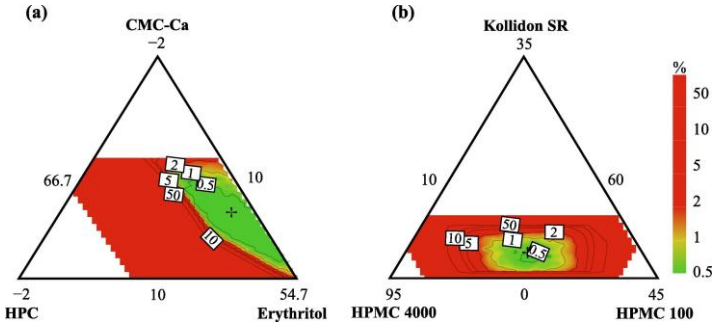
Group	Disintegration	Contact angle	Gel strength	Gelation	Water uptake	Mass loss
Lactose monohydrate	0.938	1.000	1.000	1.000	1.000	1.000
Starch 1500	0.875	0.000	0.000	0.543	0.011	0.000
MCC 102	0.594	0.000	0.133	1.000	0.611	0.037
D-mannitol	0.625	0.000	1.000	0.509	0.412	0.017

Normality test results

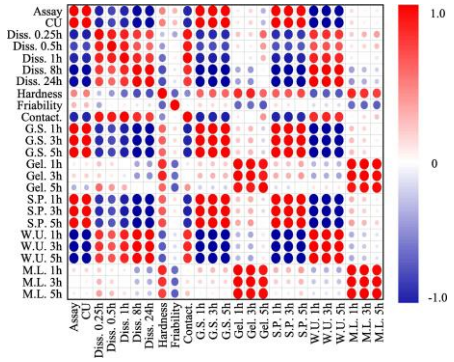
Group	Kolmogorov-Smirnov	Shapiro-Wilk
Contact angle		
Sarpogrelate HCl	0.102	0.092
Starch 1500	0.101	0.058
MCC 102	0.084	0.054
D-mannitol	0.071	0.058
Lactose monohydrate	0.080	0.059
Gel strength		
Sarpogrelate HCl	0.200*	0.223
Starch 1500	0.200*	0.749
MCC 102	0.200*	0.824
D-mannitol	0.100	0.186
Lactose monohydrate	0.200*	0.334
Gelation		
Sarpogrelate HCl	0.200*	0.114
Starch 1500	0.200*	0.606
MCC 102	0.200*	0.538
D-mannitol	0.199	0.224
Lactose monohydrate	0.200*	0.179
Water uptake		
Sarpogrelate HCl	0.113	0.072
Starch 1500	0.200*	0.695
MCC 102	0.053	0.083
D-mannitol	0.200*	0.207
Lactose monohydrate	0.111	0.076
Mass loss		
Sarpogrelate HCl	0.200*	0.375
Starch 1500	0.200*	0.425
MCC 102	0.200*	0.449
D-mannitol	0.200*	0.658
Lactose monohydrate	0.200*	0.574

(2) 위약층을 활용한 QbD 기반 처방 최적화

- 각 층(IR, SR)에 대한 처방 최적화 후
  Monte Carlo simulation을 활용한 설계공간 도출

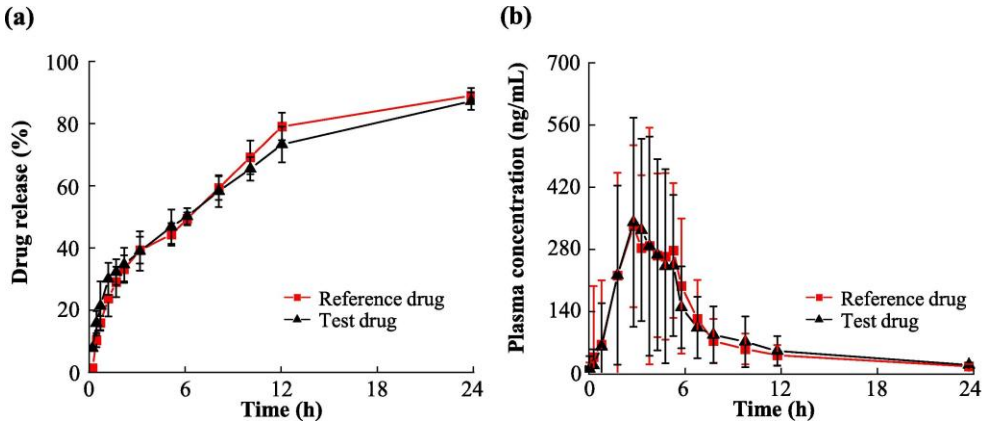


- 다변량 분석을 활용한
  반제품-완제품 상관관계 도출



(3) 최적처방과 대조약 생물학적 동등성 평가

- 최적처방과 대조약에 대한 In vitro (f2 value) 및 In vivo (임상) 평가



CONTENT.3

R&D Projects

국가 및 용역과제

03

- 삼제 복합제(FDC) 정제 제조 공정 개발(용역과제)
- 점안렌즈를 이용한 약물전달 응용기술 개발(국가과제)

R&D Projects 01

과제 요약

PAT 및 딥러닝 기반 통합 공정 제어 전략을 활용한 삼제 복합제(FDC) 정제 제조 공정 개발

용역과제

참여기간: 2024.08.01-2025.12.31
소속기관: 케이아이바이오 (주)

실험목표 QbD 기반 공정 설계와 PAT 및 딥러닝 모델을 통합하여 삼제 복합제(FDC) 정제 제조 공정에서 품질 예측 및 in-process control 전략 구축

- ☑ 고전단 습식과립 및 롤러컴팩션 공정 최적화를 통한 Design Space 확립
- ☑ QA–CQA 상관관계 분석을 통한 품질 영향 인자 규명
- ☑ 딥러닝 기반 용출 예측 모델 개발을 통한 비선형 품질 예측 구현
- ☑ 공정 별 in-process control 전략 구축

■ 수행 업무 및 성과

- QbD 기반 **처방 및 공정 최적화**
- 다변량 분석 기반 공정 변수(CPP)와 품질(QA, CQA) 간 **상관관계 분석**
- 다양한 PAT 도구 (Eyecon™, Torque, DIC)를 활용한 **in-process control 전략 확립**
- ANN 기반 용출 **예측 모델 구축**
- DIC를 활용해 리본 내부 구조 불균일성 규명
- **CTD 3.2.P. 작성**

- SCIE 학술지 논문 게재 (주저자)
- 딥러닝(ANN) 기반 용출 예측 모델 구축
- 복합제 제조 공정의 재현성 및 품질 일관성 확보를 위한 QbD–PAT–AI 통합 공정 제어 전략 제시

Performance

SCIE 학술지 주저자 논문

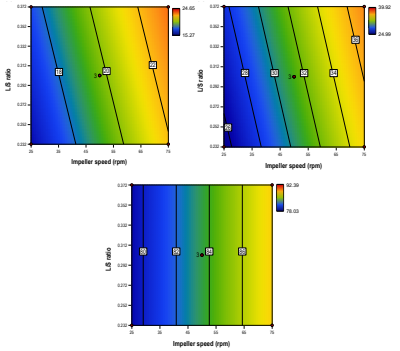
Integrated control strategy based on deep learning and process analytical technology for the manufacture of a triple fixed-dose combination tablet *Journal of Pharmaceutical Investigation, 1-21*, (2025) | 1저자

R&D Projects 01

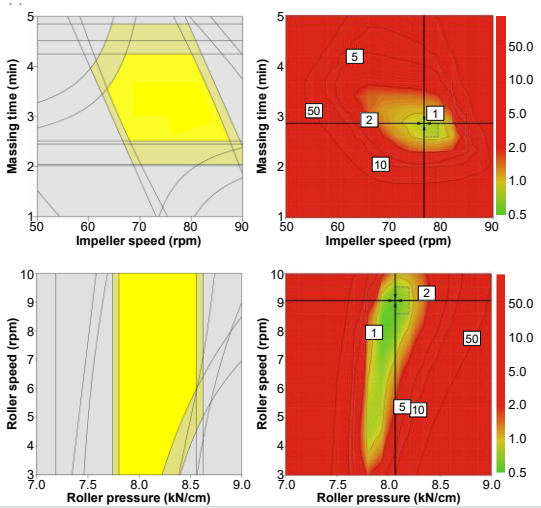
주요 연구내용

(1) 제조공정 최적화

- DoE(Box-Behnken design)을 사용한 실험수행

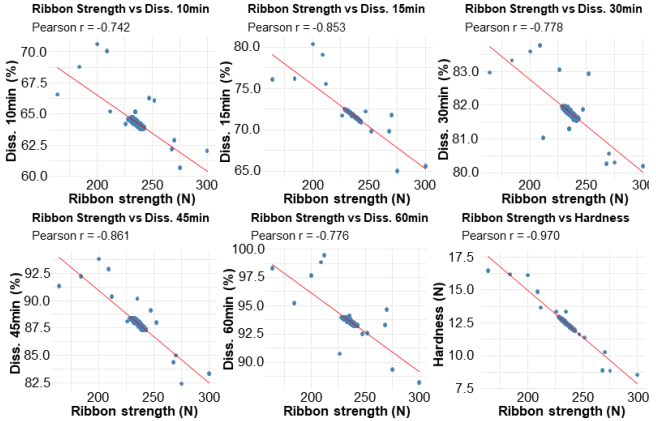


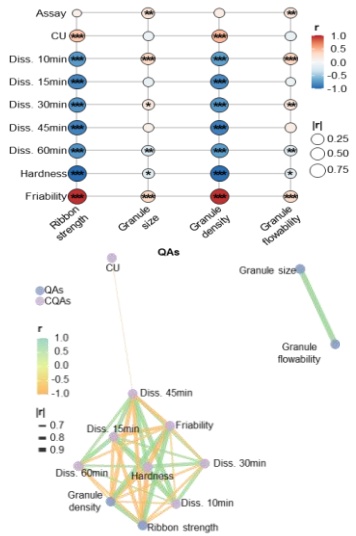
- 설계공간 도출



(2) 다변량 분석 활용 반제품-완제품 상관관계 도출

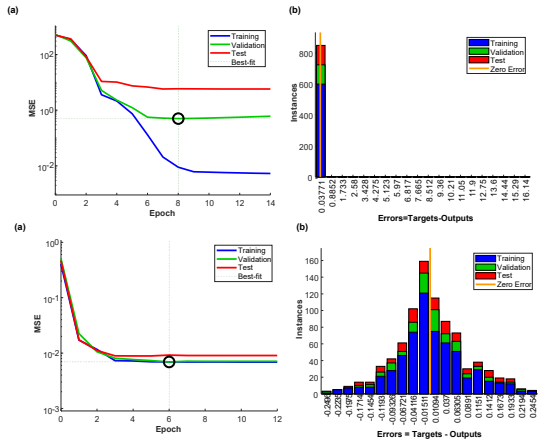
- 반제품-완제품 품질특성 간 상관관계 도출

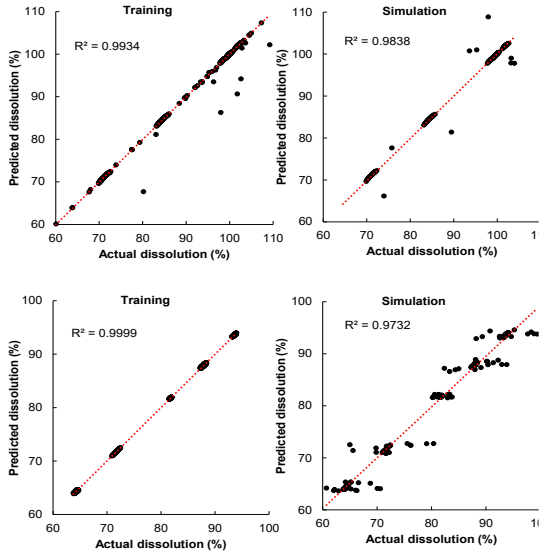




(3) 딥러닝 기반 용출 예측 모델 개발

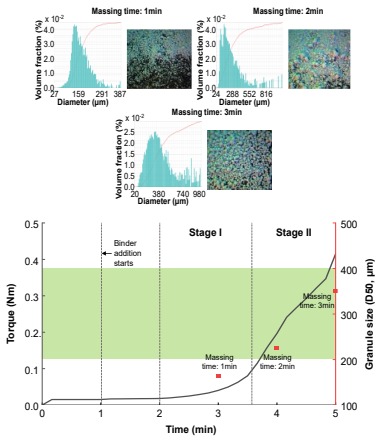
- 딥러닝으로써 ANN을 사용한 예측 모델



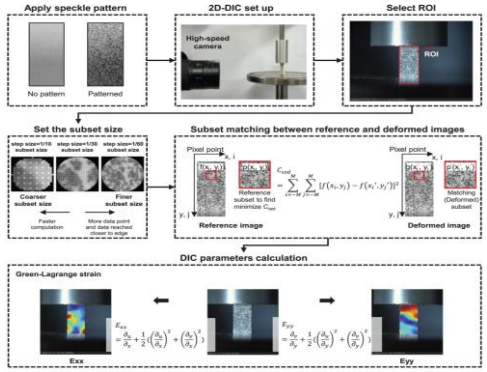


(4) Advanced PAT 활용 공정 별 in-process control 전략 구축

- 습식과립화 공정: 3D 이미징, 토크 프로파일



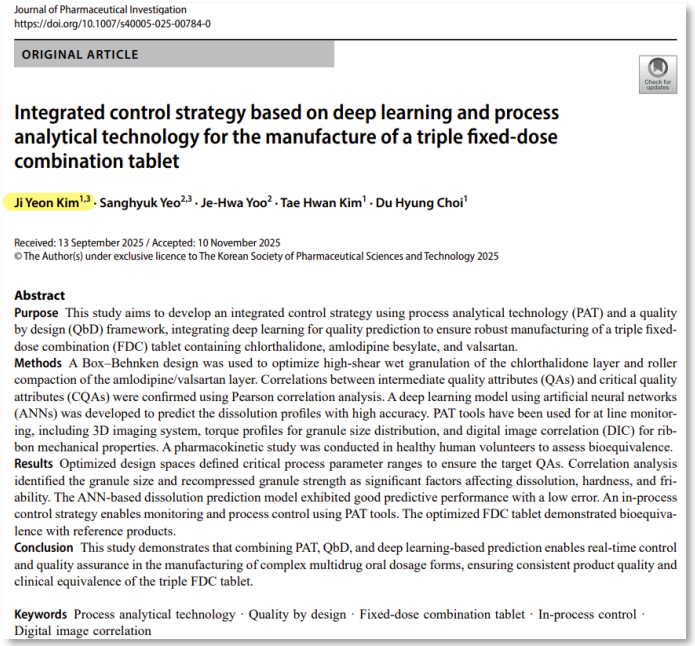
- 건식과립화 공정: 디지털 이미지 상관분석 (DIC: Digital Image Correlation)



R&D Projects 01

주요 성과요약

SCIE 학술지 논문 게재(주저자)



Journal of Pharmaceutical Investigation
https://doi.org/10.1007/s40005-025-00784-0

ORIGINAL ARTICLE

Integrated control strategy based on deep learning and process analytical technology for the manufacture of a triple fixed-dose combination tablet

Ji Yeon Kim[1,3] · Sanghyuk Yeo[2,3] · Je-Hwa Yoo[2] · Tae Hwan Kim[1] · Du Hyung Choi[1]

Received: 13 September 2025 / Accepted: 10 November 2025
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Abstract

Purpose This study aims to develop an integrated control strategy using process analytical technology (PAT) and a quality by design (QbD) framework, integrating deep learning for quality prediction to ensure robust manufacturing of a triple fixed-dose combination (FDC) tablet containing chlorthalidone, amlodipine besylate, and valsartan.

Methods A Box–Behnken design was used to optimize high-shear wet granulation of the chlorthalidone layer and roller compaction of the amlodipine/valsartan layer. Correlations between intermediate quality attributes (QAs) and critical quality attributes (CQAs) were confirmed using Pearson correlation analysis. A deep learning model using artificial neural networks (ANNs) was developed to predict the dissolution profiles with high accuracy. PAT tools have been used for at line monitoring, including 3D imaging system, torque profiles for granule size distribution, and digital image correlation (DIC) for ribbon mechanical properties. A pharmacokinetic study was conducted in healthy human volunteers to assess bioequivalence.

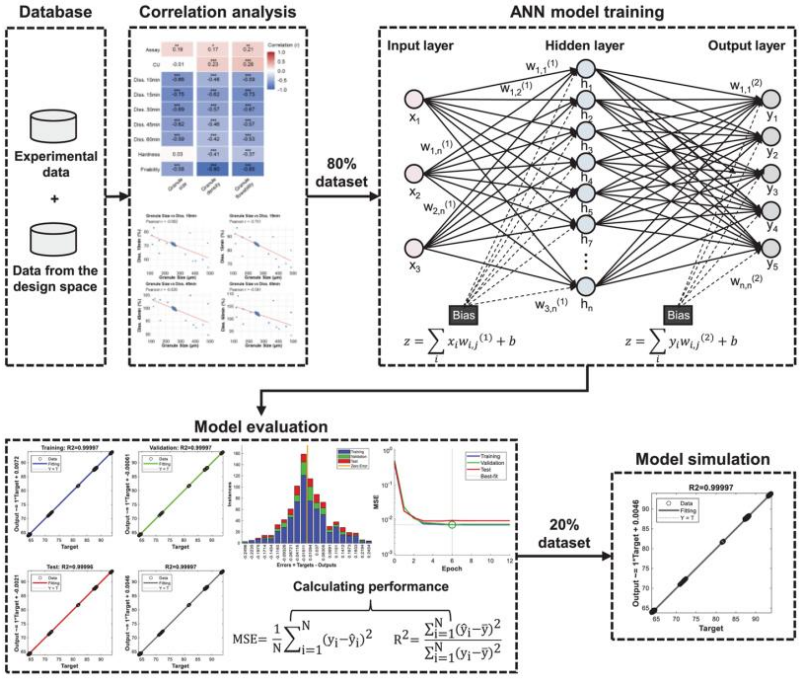
Results Optimized design spaces defined critical process parameter ranges to ensure the target QAs. Correlation analysis identified the granule size and recompressed granule strength as significant factors affecting dissolution, hardness, and friability. The ANN-based dissolution prediction model exhibited good predictive performance with a low error. An in-process control strategy enables monitoring and process control using PAT tools. The optimized FDC tablet demonstrated bioequivalence with reference products.

Conclusion This study demonstrates that combining PAT, QbD, and deep learning-based prediction enables real-time control and quality assurance in the manufacturing of complex multidrug oral dosage forms, ensuring consistent product quality and clinical equivalence of the triple FDC tablet.

Keywords Process analytical technology · Quality by design · Fixed-dose combination tablet · In-process control · Digital image correlation

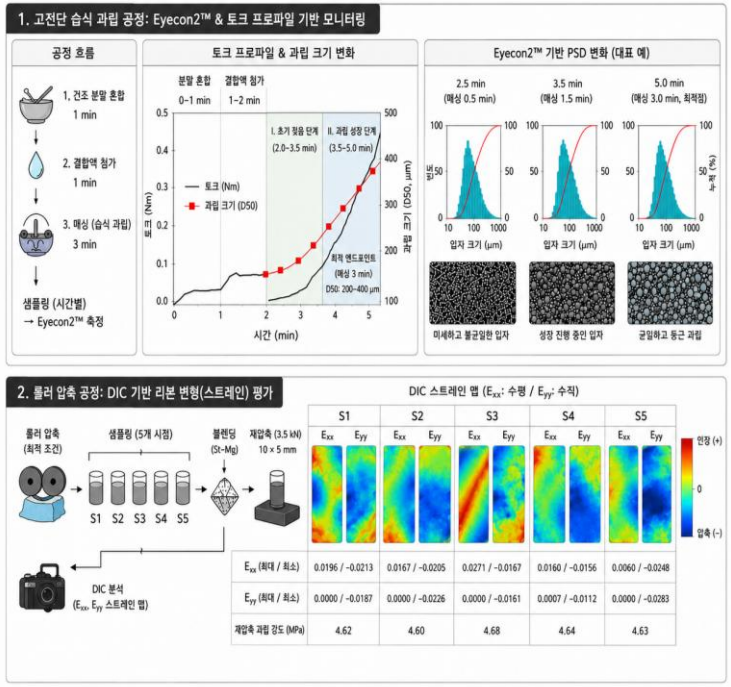
- 3제 복합제(FDC)의 고도화된 품질 관리를 위한 **PAT-QbD 통합 제어 전략** 수립
- 실시간 품질 보증(Real-time Release Testing) 가능성 입증

용출 예측 모델 구축



- 다변량 분석으로써 Pearson 상관 분석을 통한 **반제품-완제품 상관관계 도출**
- ANN(인공신경망) 모델을 활용한 비선형적 메커니즘 규명 및 고정밀 **예측 모델링 구현**

통합 공정 제어 전략 제시



- 상관분석 기반 **In-process control 전략** 수립으로 품질 관리 체계 정립
- 단위공정별 실시간 모니터링 체계 구축을 위한 **advanced PAT(3D Imaging, Torque, DIC) 사용**

🔑 핵심 수행역량

첨단 PAT(3D Imaging, Torque, DIC)와 QbD를 융합하여 복합제 공정의 변동성을 제어하고 제품 품질을 보증하는 제약 공정 설계

R&D Projects 02

과제 요약

안구건조증 치료약물의 흡수율을 증가시키기 위한 점안렌즈를 이용한 약물전달 응용기술 개발

산업통상자원부

참여기간: 2020.04.01-2024.12.31
소속기관: 대구가톨릭대학교 약학대학

실험목표 안구건조증 치료제의 흡수율 증가를 위한 약물전달 응용기술 개발

- ☑ 충전액(약물 함유 점안액) 처방 최적화
- ☑ 완제의약품(충전액 주입된 렌즈) 물리화학적 특성 평가 및 최적화
- ☑ 충전액 제조공정 Scale-up 연구

■ 수행 업무 및 성과

- QbD 접근법 기반 다양한 폴리머를 사용한 **충전액 처방 최적화**
- 완제의약품 **물리화학적 특성(산소투과도, 광투과율, 영률) 평가**
- 완제의약품 **약물 방출 평가**
- 완제의약품 약물 방출 평가를 위한 **실험 보조 도구 제작**
- 충전액 제조공정 최적화 및 전산유체역학(CFD) 시뮬레이션을 활용한 **Scale-up 연구**
- 완제의약품 품질평가를 위한 **기준 및 시험방법 작성**
- 연차별 **보고서 작성**

- 국제 학술회의(AAPS) 논문 발표(주저자)
- 완제의약품 약물 방출 평가 보조 도구 설계 및 제작
- 무차원 수 및 CFD를 활용한 scale-up 연구

Performance

국제 학술회의 주저자 논문 발표

Development of a Novel Therapeutic Contact Lens with the Embedded-Drug Chamber as an Ophthalmic Drug Delivery System for Dry Eye Syndrome *AAPS* (2023) | 1저자

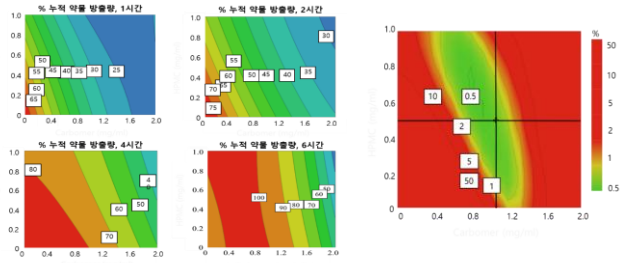
약물방출 평가 보조도구 제작

R&D Projects 02

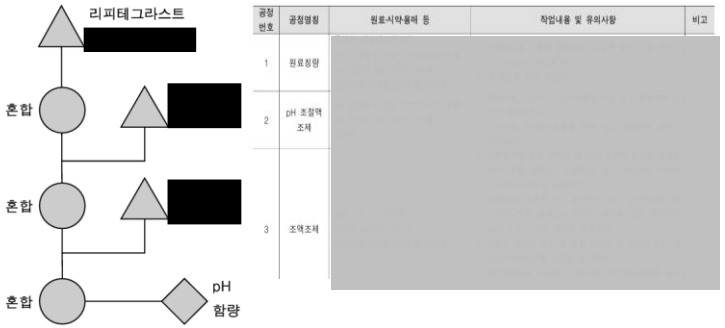
주요 연구내용

(1) 충전액(약물 함유 점안액) 처방 최적화

- QbD 기반 충전액 처방 최적화



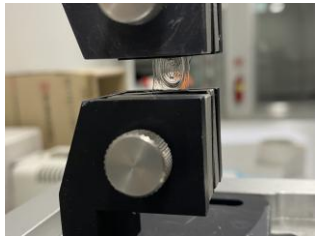
- 충전액 제조방법 및 기시법 작성

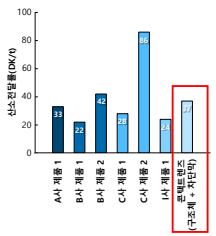


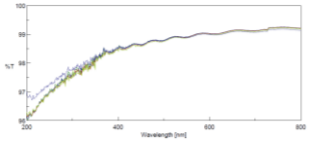


(2) 완제의약품 물리화학적 특성 평가

- 콘택트렌즈의 물리화학적 특성 평가

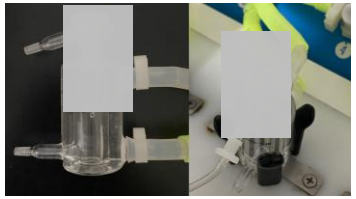


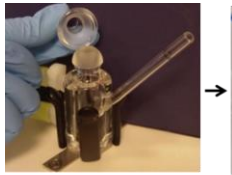




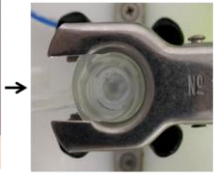
Young's modulus(Mpa) = $\frac{Stress}{Strain}$

- 콘택트렌즈의 함량, 약물방출, 안구 투과능 평가

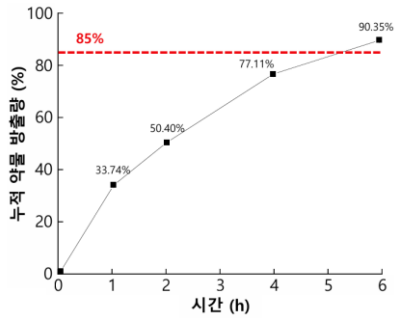


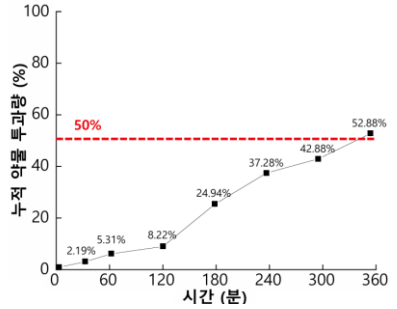


코니아 셀 (Cornea diffusion cell)



코니아 셀에 장착된 토끼각막





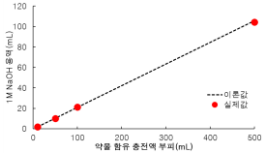
(3) 충전액 제조공정 Scale-up

- Froude number 기반 생산규모 확대에 따른 혼합속도 환산

$$Fr = \frac{R \times w^2}{g}$$

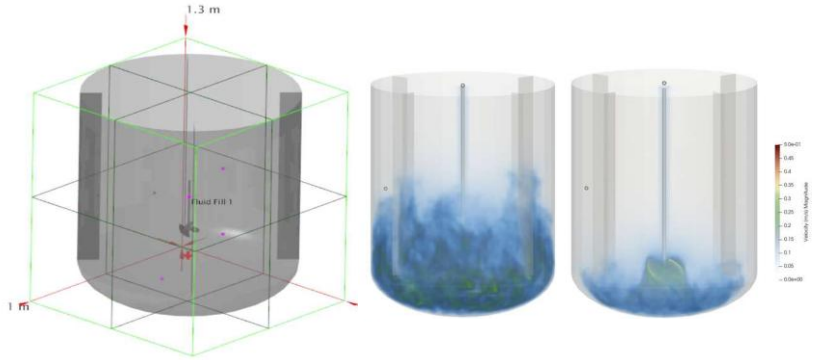
R: 혼합 챔버의 반지름(m), w: 각속도(Angular velocity, r/s), g: 중력계수

	10mL	50mL	100mL	500mL	5000mL
기하학적 특성					
혼합기 지름(mm)					
임펠러 지름(mm)					
실험 설정					
임펠러 회전수(rpm)					
Froude number					



약물 함유 충전액 부피(mL)	1M NaOH 용액(mL)		
	이론값(mL)	실제값(mL)	이론값-실제값(mL)
10			
50			
100			
500			

- 전산유체역학(CFD) 시뮬레이션을 활용한 Scale-up

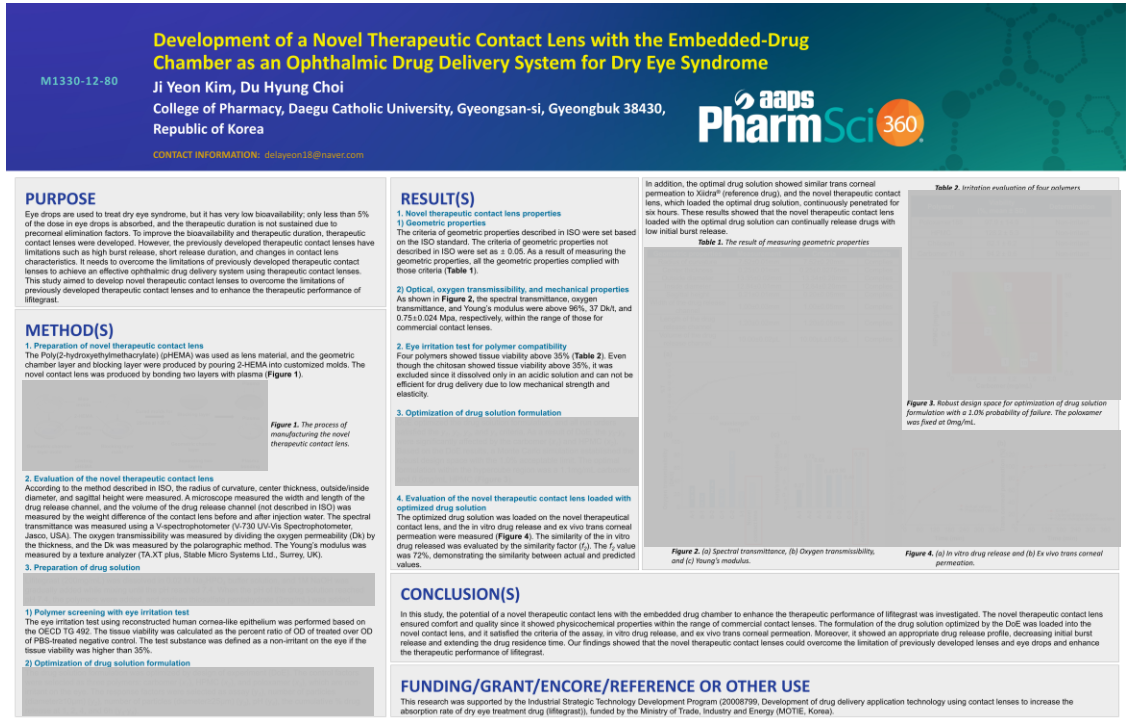


특허, 논문(작업 중) 등으로 인해 연구 결과 및 상세 내용 공개 불가

R&D Projects 02

주요 성과요약

국제 학술회의의 주저자 논문 발표



Development of a Novel Therapeutic Contact Lens with the Embedded-Drug Chamber as an Ophthalmic Drug Delivery System for Dry Eye Syndrome

M1330-12-80

Ji Yeon Kim, Du Hyung Choi
College of Pharmacy, Daegu Catholic University, Gyeongsan-si, Gyeongbuk 38430, Republic of Korea
CONTACT INFORMATION: delayeon18@naver.com

aaps PharmSci 360

PURPOSE

Eye drops are used to treat dry eye syndrome, but it has very low bioavailability; only less than 5% of the dose in eye drops is absorbed, and the therapeutic duration is not sustained due to precorneal elimination factors. To improve the bioavailability and therapeutic duration, therapeutic contact lenses were developed. However, the previously developed therapeutic contact lenses have limitations such as high burst release, short release duration, and changes in contact lens characteristics. It needs to overcome the limitations of previously developed therapeutic contact lenses to achieve an effective ophthalmic drug delivery system using therapeutic contact lenses. This study aimed to develop novel therapeutic contact lenses to overcome the limitations of previously developed therapeutic contact lenses and to enhance the therapeutic performance of liftegrast.

METHOD(S)

1. Preparation of novel therapeutic contact lens
The Poly(2-hydroxyethylmethacrylate) (pHEMA) was used as lens material, and the geometric chamber layer and blocking layer were produced by pouring 2-HEMA into customized molds. The novel contact lens was produced by bonding two layers with plasma (**Figure 1**).

Figure 1. The process of manufacturing the novel therapeutic contact lens.

2. Evaluation of the novel therapeutic contact lens
According to the method described in ISO, the radius of curvature, center thickness, outside/inside diameter, and sagittal height were measured. A microscope measured the width and length of the drug release channel, and the volume of the drug release channel (not described in ISO) was measured by the weight difference of the contact lens before and after injection water. The spectral transmittance was measured using a V-spectrophotometer (V-730 UV-Vis Spectrophotometer, Jasco, USA). The oxygen transmissibility was measured by dividing the oxygen permeability (Dk) by the thickness, and the Dk was measured by the polarographic method. The Young's modulus was measured by a texture analyzer (TA.XT plus, Stable Micro Systems Ltd., Surrey, UK).

3. Preparation of drug solution

1) Polymer screening with eye irritation test
The eye irritation test using reconstructed human cornea-like epithelium was performed based on the OECD TG 492. The tissue viability was calculated as the percent ratio of OD of treated over OD of PBS-treated negative control. The test substance was defined as a non-irritant on the eye if the tissue viability was higher than 35%.

2) Optimization of drug solution formulation

RESULT(S)

1. Novel therapeutic contact lens properties
1) Geometric properties
The criteria of geometric properties described in ISO were set based on the ISO standard. The criteria of geometric properties not described in ISO were set as ± 0.05. As a result of measuring the geometric properties, all the geometric properties complied with those criteria (**Table 1**).

2) Optical, oxygen transmissibility, and mechanical properties
As shown in **Figure 2**, the spectral transmittance, oxygen transmittance, and Young's modulus were above 96%, 37 Dk/t, and 0.75±0.024 Mpa, respectively, within the range of those for commercial contact lenses.

2. Eye irritation test for polymer compatibility
Four polymers showed tissue viability above 35% (**Table 2**). Even though the chitosan showed tissue viability above 35%, it was excluded since it dissolved only in an acidic solution and can not be efficient for drug delivery due to low mechanical strength and elasticity.

3. Optimization of drug solution formulation

4. Evaluation of the novel therapeutic contact lens loaded with optimized drug solution
The optimized drug solution was loaded on the novel therapeutic contact lens, and the in vitro drug release and ex vivo trans corneal permeation were measured (**Figure 4**). The similarity of the in vitro drug released was evaluated by the similarity factor (f_2). The f_2 value was 72%, demonstrating the similarity between actual and predicted values.

In addition, the optimal drug solution showed similar trans corneal permeation to Xiidra® (reference drug), and the novel therapeutic contact lens, which loaded the optimal drug solution, continuously penetrated for six hours. These results showed that the novel therapeutic contact lens loaded with the optimal drug solution can continually release drugs with low initial burst release.

Table 1. The result of measuring geometric properties

Table 2. Irritation evaluation of four polymers

Figure 3. Robust design space for optimization of drug solution formulation with a 2.0% probability of failure. The poloxamer was fixed at 0mg/mL.

Figure 2. (a) Spectral transmittance, (b) Oxygen transmissibility, and (c) Young's modulus.

Figure 4. (a) In vitro drug release and (b) Ex vivo trans corneal permeation.

CONCLUSION(S)

In this study, the potential of a novel therapeutic contact lens with the embedded drug chamber to enhance the therapeutic performance of liftegrast was investigated. The novel therapeutic contact lens ensured comfort and quality since it showed physicochemical properties within the range of commercial contact lenses. The formulation of the drug solution optimized by the DoE was loaded into the novel contact lens, and it satisfied the criteria of the assay, in vitro drug release, and ex vivo trans corneal permeation. Moreover, it showed an appropriate drug release profile, decreasing initial burst release and extending the drug residence time. Our findings showed that the novel therapeutic contact lenses could overcome the limitation of previously developed lenses and eye drops and enhance the therapeutic performance of liftegrast.

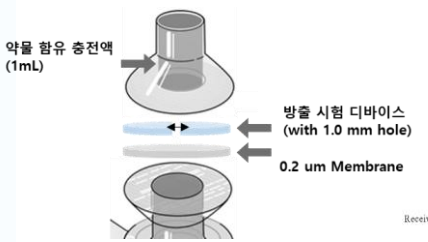
FUNDING/GRANT/ENCORE/REFERENCE OR OTHER USE

This research was supported by the Industrial Strategic Technology Development Program (20008799, Development of drug delivery application technology using contact lenses to increase the absorption rate of dry eye treatment drug (liftegrast)), funded by the Ministry of Trade, Industry and Energy (MOTIE, Korea).

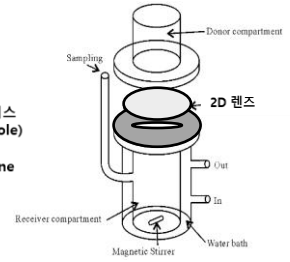
– 약물 저장소를 갖춘 콘택트렌즈 시스템에 최적화된 **충전액(약물 함유 점안액) 처방 설계** 및 약물 흡수율 증대 효과 입증 (AAPS 주저자 발표)

약물방출 평가 보조도구 제작

[기존 약물 방출 평가 방법]



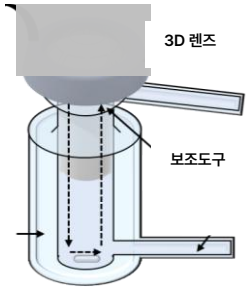
[충전액 평가 방법]



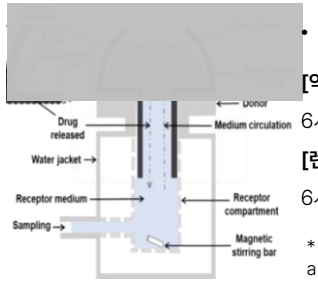
[2D 렌즈 평가 방법]

- 점안액 또는 2D 형상의 렌즈만 평가 가능
- 3D 형상의 렌즈의 경우 렌즈 중앙이 눌려 **정확한 약물 방출 평가가 어려움**
- 렌즈에서 방출된 약물이 다시 렌즈에 흡착되어 **불균일한 약물 방출 패턴**을 나타냄

렌즈 구조를 유지한 채 평가 가능한 평가 방법 개발 필요



[보조도구 장착]



[보조 도구의 원리]

- 렌즈의 형상을 유지하며 약물 방출 평가 가능한 보조도구 설계

[약물 방출률]*
6시간 30% 미만[a] → 89.6%

[렌즈 간 편차]
6시간 30% 미만 → 89.6%

* 기존 방법과 동일 배치의 렌즈 평가 결과
a 약물 방출 시험 후 렌즈를 회수하여 함량 평가 결과 95% 이상으로 렌즈 내에서 약물이 방출되지 않음을 확인

– 최적화된 충전액 처방의 방출 거동을 정밀 평가하기 위해 렌즈 구조를 반영한 **전용 In-vitro 평가 시스템 개발**

특허, 논문(작업 중) 등으로 인해 연구 결과 및 상세 내용 공개 불가

🔑 핵심 수행역량

기존 평가법의 한계를 극복하기 위해 자체 시험 시스템을 설계·구축하고, 특수 제형의 약물 방출 메커니즘 규명

Research Publications & Projects

Publications 2021–2026 | SCI(E) 논문 10편 (1저자 9편)

연도	게재지/논문 제목	역할
2026	European Journal of Pharmaceutical Sciences | (최종 출판 전 편집 중) **Development and optimization of an LED–based particle image velocimetry methodology for dynamic powder flowability in pharmaceutical manufacturing**	1저자
2025	Journal of Pharmaceutical Investigation,1-21, 2025 | **Integrated control strategy based on deep learning and process analytical technology for the manufacture of a triple fixed-dose combination tablet**	1저자
2025	International Journal of Pharmaceutics, 678, 125690, 2025 | **Deep learning-based image classification and quantification models for tablet sticking**	1저자
2023	Journal of Pharmaceutical Investigation, 53(6), 881-894, 2023 | **Developmental strategy for swellable/erodible matrix tablet of mirabegron: quality by design approach with various geometric properties and pharmacokinetic evaluation**	1저자
2023	Journal of Pharmaceutical Investigation, 53(5), 707-722, 2023 | **Development of prediction model with machine learning in continuous twin screw granulation**	1저자
2022	Journal of Pharmaceutical Investigation, 53(3), 389-406, 2022 | **Quality by design approach with multivariate analysis and artificial neural network models to understand and control excipient variability**	1저자
2022	Pharmaceutics, 14(11), 2416, 2022 | **Control Strategy for Excipient Variability in the Quality by Design Approach Using Statistical Analysis and Predictive Model: Effect of Microcrystalline Cellulose Variability on Design Space**	1저자
2022	International Journal of Pharmaceutics, 618, 121659, 2022 | **A formulation development strategy for dual-release bilayer tablets: An integrated approach of quality by design and a placebo layer**	1저자
2021	Pharmaceutics, 13(9), 1443, 2021 | **Development of a robust control strategy for fixed-dose combination bilayer tablets with integrated quality by design, statistical, and process analytical technology approach**	2저자
2021	Pharmaceutics, 13(1), 80, 2021 | **Control strategy for process development of high-shear wet granulation and roller compaction to prepare a combination drug using integrated quality by design**	1저자

Research Publications & Projects

R&D Projects (국가 연구과제) 2020–2025 | 국가 R&D 연구과제 10건 참여 | 국내 제약회사 용역과제 참여

기관	과제명	참여 기간
산업통상자원부	고품질 설계기반 고혈압 치료제 생산용 핵심 소재 및 중간체, 완제 생산 공정 기술개발	25.03 - 25.12
산업통상자원부	항체의약품의 장내 수송체 표적형 경구 약물전달시스템 및 제형화 기술개발	24.06 - 25.02
산업통상자원부	안구건조증 치료 약물(lifitegrast)의 흡수율을 증가시키기 위한 점안렌즈를 이용한 약물전달 응용기술 개발	20.04 - 24.12
연구개발특구진흥재단	풍선효과를 이용한 지속적 약물 방출 캡슐 개발을 위한 시제품 제작 연구	23.06 - 24.11
식품의약품안전처	글로벌GMP기반 제약혁신기술 도입- 의약품 품질시스템 기반 예시모델 및 기초기술 개발, 연속공정 적용을 위한 원료물질특성(Materials Characterisation)과 연속공정 중 반제품 품질 관리전략(Control Strategy) 개발과 적용 방법에 대한 연구	24.03 - 24.11
식품의약품안전처	의약품 설계기반 품질고도화(QbD) 제도 도입기반 구축: 기초기술 개발, 다양한 공정분석기술(PAT) 특성과 적용 방법	23.04 - 23.11
한국연구재단	메커니즘기반 모델을 이용한 다층복합제형 제조공정 변동성이 품질에 미치는 물리화학적 기전 연구	21.03 - 23.05
연구개발특구진흥재단	OAB 치료 서방형 신제품 개발을 위한 시제품 제작 연구	22.06 - 23.05
식품의약품안전처	의약품 설계기반 품질고도화(QbD) 제도 도입기반 구축: 기초기술 개발, QbD 접근방식에서의 연속공정 방법론 개발	22.04 - 22.11
식품의약품안전처	의약품 설계기반 품질고도화(QbD) 제도 도입기반 구축: 기초기술 개발, QbD 기반 지식관리 방법론 개발	21.04 - 21.11

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